

RARITONE – AI POWERED VIRTUAL TRY-ON
PLATFORM

Internship Project Report

Submitted By
Vagdevi Malineni

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DECLARATION

I hereby declare that the project report entitled “**Raritone – AI Powered Virtual Try-On Platform**” submitted in partial fulfillment of the internship requirements is my original work.

The information and results presented in this report are based on the work carried out during the internship period and have not been submitted elsewhere for any academic award.

Signature: Vagdevi Malineni

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I express my sincere gratitude to the management and technical team of RARITONE for providing me with the opportunity to work on this project.

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ABSTRACT

The rapid growth of online fashion retail has increased the demand for personalized shopping experiences. Traditional e-commerce platforms often suffer from challenges such as inaccurate size selection, poor product visualization, and lack of personalization, leading to customer dissatisfaction and increased product returns.

The **Raritone – AI Powered Virtual Try-On Platform** aims to overcome these challenges by providing a secure, scalable, and AI-ready backend infrastructure that supports avatar management, body measurements, product mapping, and virtual try-on workflows.

The backend was developed using **Node.js**, **Express.js**, and **MongoDB Atlas**, while **Cloudinary** was integrated for media storage and optimization. Security mechanisms such as JWT Authentication, Role-Based Access Control, Joi Validation, Helmet, and Rate Limiting were implemented to ensure secure API communication.

The architecture has been designed to support future AI integration for body analysis, garment fitting, fit score generation, and intelligent size recommendations.

The project successfully establishes a modern backend foundation capable of supporting next-generation virtual fashion technologies.

TABLE OF CONTENTS

1. Introduction
2. Problem Statement
3. Objectives
4. Technology Stack
5. System Architecture
6. Database Design
7. Modules Developed
8. Security Implementation
9. Media Processing Workflow
10. API Development
11. Testing
12. AI Integration Planning
13. Scalability Research
14. Challenges Faced
15. Key Learnings
16. Future Enhancements
17. Conclusion

1. INTRODUCTION

The online fashion industry faces challenges such as inaccurate size selection, uncertainty regarding garment appearance, and lack of personalized shopping experiences.

Raritone addresses these challenges through an AI-powered virtual try-on platform that enables users to create digital avatars, store body measurements, receive personalized recommendations, and virtually visualize products before purchase.

The backend serves as the core foundation of the platform by managing user information, measurements, product data, security, and future AI services.

2. PROBLEM STATEMENT

Traditional online shopping platforms suffer from several limitations:

- Customers cannot visualize products before purchasing.
- Size recommendations are often inaccurate.
- Product return rates remain high.
- Personalized fashion recommendations are limited.
- Existing systems lack intelligent fitting mechanisms.

The project aims to build a scalable backend architecture capable of supporting future AI-powered virtual try-on experiences.

3. OBJECTIVES

The primary objectives of the project are:

- Develop secure backend APIs.
- Implement authentication and authorization.
- Design scalable database schemas.
- Manage user avatars and measurements.
- Support virtual try-on workflows.
- Prepare infrastructure for AI integration.
- Ensure security, performance, and maintainability.
- Build a modular architecture for future expansion.

4. TECHNOLOGY STACK

Backend Technologies

- Node.js
- Express.js

Database

- MongoDB Atlas
- Mongoose ODM

Authentication

- JWT Authentication
- Role-Based Access Control

Media Handling

- Multer
- Sharp

- Cloudinary

Security

- Helmet
- Express Rate Limiting
- CORS

Logging & Monitoring

- Morgan
- Sentry
- Grafana
- Firebase Analytics

Future Technologies Researched

- Redis
- Docker
- Load Balancers
- Microservices
- Queue Systems

5. SYSTEM ARCHITECTURE

The backend follows a modular layered architecture.

Frontend / Client



REST API Layer

↓

Authentication Layer

↓

Business Logic Layer

↓

MongoDB Atlas

↓

Clouinary Storage

↓

AI Services (Future)

Advantages

- Easy maintenance
- Better scalability
- Improved security
- Modular development
- AI readiness

6. DATABASE DESIGN

MongoDB Atlas was used as the primary database.

Collections Implemented

1. Users
2. Products
3. Avatars

4. Measurements
5. TryOns
6. ProductMappings

Relationships

User → Avatar

User → Measurements

User → TryOns

TryOn → Product

Product → ProductMapping

Benefits

- Scalable schema design
- Faster querying
- AI-ready structure
- Easy future expansion

7. MODULES DEVELOPED

Authentication Module

Features

- User Registration
- User Login
- JWT Authentication
- Role Authorization

Benefits

- Secure user management
- Protected APIs

Product Module

Features

- Product Creation
- Product Updates
- Product Search
- Product Filtering

Benefits

- Efficient catalog management
- Product organization

Avatar Module

Features

- Avatar Upload
- Cloud Storage
- Image Optimization

Benefits

- Personalized digital identity
- Virtual try-on support

Measurement Module

Features

- Body Measurement Storage
- User Measurement Profiles

Benefits

- Accurate size recommendations
- Reduced return rates

Product Mapping Module

Features

- Product Relationships
- Recommendation Support

Benefits

- Smart product matching
- AI compatibility

Try-On Module

Features

- Virtual Try-On Requests
- Product Association

Benefits

- Better shopping experience

- Future AI integration support

8. SECURITY IMPLEMENTATION

The backend includes multiple security layers.

JWT Authentication

Protects secured APIs.

Role-Based Access Control

Roles:

- USER
- ADMIN
- SUPER_ADMIN

Helmet

Provides security headers.

Rate Limiting

Protects against abuse and excessive requests.

Joi Validation

Validates incoming data.

Error Middleware

Centralized error management.

9. MEDIA PROCESSING WORKFLOW

User Upload



Multer Upload



Sharp Compression



Cloudinary Upload



CDN Optimization



Database Storage



Response

Benefits

- Faster image loading
- Reduced storage consumption
- Optimized media delivery

10. API DEVELOPMENT

REST APIs were developed for:

- Authentication
- Products
- Avatars

- Measurements
- Product Mapping
- Try-On

Features:

- JWT Protected Routes
- Validation Middleware
- Error Handling
- Structured Responses

11. TESTING

The backend was tested using Postman.

Testing Areas

- Authentication
- CRUD Operations
- Protected Routes
- Validation
- Error Handling
- Database Integration

Results

All modules operated successfully without critical issues.

12. AI INTEGRATION PLANNING

The architecture was designed to support future AI services.

Planned Features

- AI Avatar Generation
- Body Analysis
- Garment Matching
- Virtual Fitting
- Size Recommendation
- Fit Score Prediction

Benefits

- Improved shopping experience
- Reduced returns
- Personalized recommendations

13. SCALABILITY RESEARCH

Several scalability solutions were researched.

Redis

Caching frequently accessed data.

Load Balancing

Handling increased user traffic.

Docker

Containerized deployment.

Microservices

Independent service scaling.

MongoDB Atlas Clustering

Database growth management.

14. CHALLENGES FACED

Challenge 1

Designing scalable database schemas.

Solution:

Implemented modular MongoDB collections.

Challenge 2

Securing APIs.

Solution:

Used JWT, RBAC, Helmet, and Validation.

Challenge 3

Image handling and storage.

Solution:

Integrated Multer, Sharp, and Cloudinary.

Challenge 4

Preparing for AI integration.

Solution:

Created AI-ready workflows and schemas.

15. KEY LEARNINGS

Throughout the internship, significant knowledge was gained in:

- Node.js Development
- Express.js Framework
- MongoDB Design
- REST API Development
- JWT Authentication
- Security Best Practices
- Cloud Storage Integration
- Database Relationships
- Documentation Writing
- Git & GitHub Collaboration
- Team-Based Development

16. FUTURE ENHANCEMENTS

Planned improvements include:

- Full AI Virtual Try-On
- Automated Size Prediction
- Recommendation Engine
- Redis Integration
- Docker Deployment

- Mobile Application Support
- Advanced Analytics Dashboard
- Real-Time Notifications
- Microservices Architecture

17. CONCLUSION

The Raritone – AI Powered Virtual Try-On Platform successfully established a secure, scalable, and AI-ready backend foundation for a modern virtual fashion ecosystem.

The project implemented essential backend functionalities including authentication, avatar management, measurement storage, product mapping, media handling, security mechanisms, and scalable database design.

The internship provided valuable exposure to real-world backend development practices, software architecture, database management, cloud integration, and API development, strengthening both technical and professional skills required in modern software engineering.